

User Guide

Context

An OVA is a virtual machine distribution format that encapsulates the disk image, configuration, and metadata into a single file. This allows the same environment to be imported and executed on different machines, ensuring reproducibility. The use of OVAs for running GenESyS in laboratory environments is justified by the lack of administrator privileges on laboratory computers, which requires the use of a virtualization system in order to install the necessary dependencies.

Usage

The OVA supports the use of GenESyS as a regular user through the application available in the start menu/development, which always corresponds to the latest release available in the Git repository. For developers, the *QtCreator IDE* is also provided, it is located in the same place as the user application, along with the option to keep the repository located at `/home/vboxuser/Documents` automatically updated using either the *currentStable* or *WorkInProgress* branch.

To compile and run the development branch of the program using QtCreator, located in the Documents folder, open QtCreator and select the “Open Project” option. Then, choose the CMakeLists.txt file located in the project’s root directory. After opening the project, click “Configure Project” so that the build environment is configured automatically.

The compilation and execution target can be selected through the monitor icon button located above the “Play” (▶) button. After selecting the desired target, start the compilation and execution process by clicking the play button.

Login Credentials

Username: vboxuser

Password: 123456

Step-by-Step Guide for Installing the .ova

- Download the .ova file;
- Launch Oracle VirtualBox;
- To import the .ova, click on “Import Appliance...”;
- Select the .ova file and click “Next”, then click “Finish” on the following screen.

How to Open the Terminal

The simplest way is by using the shortcut Ctrl + Alt + T. However, the terminal is also available through the start menu under “System” > “XTerm”.

How to Increase the Terminal Font Size

Hold Ctrl and right-click inside the open terminal window.

How to Enable Guest Additions

(enables copy and paste functionality and screen resizing)

Open Oracle VirtualBox on the host machine and check its version by clicking “Help” and then “About VirtualBox...” (Example: if the version is 7.0.6r155176, consider only 7.0.6). With the version number identified, execute the following command inside the virtual machine terminal:

```
$ install_guest_add <version>
```

After the installation is complete, restart the virtual machine. Finally, enable the “Bidirectional” option available under “Devices” > “Shared Clipboard”.

Note that, in the provided terminal emulator, the paste shortcut is Shift + Insert.

How to Change Branches

To switch branches while keeping automatic updates enabled, modify the branch name in the file:

```
/home/vboxuser/.genesys_dev_branch
```

After modifying the file, restart the virtual machine for the changes to take effect.